

Mathematics for Machine Learning

Vazgen Mikayelyan



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Example

Let X be a geometric random variable with parameter p . Prove that

$$\mathbb{E}[X] = \frac{1}{p}, \text{Var}(X) = \frac{1-p}{p^2}.$$

Uniform Distribution

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A random variable is said to be Uniformly distributed on the interval (a, b) if its PDF is given by

$$f(x) = \begin{cases} 0, & \text{if } x \notin (a, b), \\ \frac{1}{b-a}, & \text{if } x \in (a, b). \end{cases}$$

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Proposition

Prove that CDF of uniform random variable on the interval (a, b) is the following function

$$F(x) = \begin{cases} 0, & \text{if } x \leq a, \\ \frac{x-a}{b-a}, & \text{if } x \in (a, b), \\ 1, & \text{if } x \geq b. \end{cases}$$

Continuous random variables

Normal Random Variable

A continuous random variable X is said to be **Normally Distributed** (or simply, **Normal Random Variable**) with parameters μ and σ^2 if the PDF of X has the following form:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < +\infty.$$

The parameter μ is called **mean**. The fact that X is a normal RV with parameters μ and σ^2 will be denoted as $X \sim N(\mu, \sigma^2)$.

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Example

Prove that if X is normally distributed with parameters μ and σ^2 , then $Y = aX + b$ is normally distributed with parameters $a\mu + b$ and $a^2\sigma^2$.

Standardization of Normal Distribution

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Example

Show that the parameters μ and σ^2 of a normal random variable represent, respectively, its expected value and variance.

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Exponential random variable

A continuous RV whose PDF is given, for some $\lambda > 0$, by

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

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Example

Let X be an exponential random variable with parameter λ . Find the CDF $F(a)$ of the RV X .

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We say that a nonnegative random variable X is **memoryless** if

$$\mathbb{P}\{X > s + t | X > t\} = \mathbb{P}\{X > s\} \text{ for all } s, t \geq 0.$$

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Example

The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda = \frac{1}{2}$. What is

(a) the probability that a repair time exceeds 2 hours?

(b) the conditional probability that a repair takes at least 10 hours, given that its duration exceeds 9 hours?

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Suppose that the number of km that a car can run before its battery wears out is exponentially distributed with an average value of 20,000 km. If a person desires to take a 1000-km trip, what is the probability that he or she will be able to complete the trip without having to replace the car battery? What can be said when the distribution is not exponential?

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$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

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- 2 $|\rho(X, Y)| \leq 1$,
- 3 $Y = aX + b$, for some constants a, b iff $\rho(X, Y) = \pm 1$.

Sequence of Functions

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We will say that f_n converges to f uniformly on X , if for all $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ holds $|f_n(x) - f(x)| < \varepsilon$ for all $x \in X$. We will write $f_n(x) \rightrightarrows f(x), x \in X$.

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Theorem

f_n converges uniformly on X if and only if for all $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0, m \in \mathbb{N}$ holds $|f_{n+m}(x) - f_n(x)| < \varepsilon$ for all $x \in X$.

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Theorem

If there exists a sequence of real numbers λ_n such that $|u_n(x)| \leq \lambda_n$ for all $x \in X$ and $n \in \mathbb{N}$ and $\sum_{n=1}^{\infty} \lambda_n < +\infty$, then $\sum_{n=1}^{\infty} u_n(x)$ is uniformly convergent.

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Example

Prove that the series $\sum_{n=1}^{\infty} \frac{\sin nx}{1+n^2}$ is uniformly convergent on $\mathbb{R}$.